MILITARY SPECIFICATION

PHENYLPROPANOLAMINE HYDROCHLORIDE, CODEINE PHOSPHATE, AND GUAIFENESIN SYRUP

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Phenylpropanolamine Hydrochloride, Codeine Phosphate, and Guaifenesin Syrup (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA).

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Applications for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Each 5 ml of syrup shall contain the following:

Phenylpropanolamine Hydrochloride	18.75 mg
Codeine Phosphate	10 mg
Guaifenesin	100 mg

3.1.1 Alcohol. The syrup shall contain 7.5 percent alcohol by volume.

3.2 Additives. The coloring, flavoring and sweetening agents, and preservatives used in the syrup shall be in the amounts approved by the FDA.

3.3 Finished product.

3.3.1 Assay.

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3.3.1.1 Phenylpropanolamine hydrochloride. The syrup shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of phenylpropanolamine hydrochloride when determined as specified in 4.7.1.

3.3.1.2 Codeine phosphate. The syrup shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of codeine phosphate when determined as specified in 4.7.1.

3.3.1.3 Guaifenesin. The syrup shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of guaifenesin when determined as specified in 4.7.1.

3.3.1.4 Alcohol content. The syrup shall assay to contain not less than 6.65 percent and not more than 8.25 percent alcohol v/v (90.0 to 110.0 percent of the required amount), when determined as specified in 4.7.2.

3.3.2 pH. The pH of the syrup shall be not less than 2.5 and not more than 4.0 when determined potentiometrically at 25°C, using the USP method.

3.3.3 Flavor and palatability. The finished syrup shall be palatable and pleasant to the taste with no unpleasant after-taste. Not later than the time specified for opening of bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) units of issue of the finished syrup, representative of the product which the offeror proposes to furnish. Two (2) samples will be subjected to panel testing for a determination of palatability (see 4.7.3 - Palatability test). The remaining samples will be used by cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determined that such product was previously procured and/or tested by the purchasing activity and found to comply with the palatability requirement.

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3.4 Ingredients.

3.4.1 Phenylpropanolamine hydrochloride and guaifenesin. The phenylpropanolamine hydrochloride and guaifenesin used in the manufacture of the finished product shall comply with the tests, standards, and requirements of the NF, including any supplements or revisions thereto.

3.4.2 Codeine phosphate. The codeine phosphate used in the manufacture of the finished product shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.5 Other ingredients. All other ingredients entering into the manufacture of the finished preparation shall be of USP or NF quality, or if not included in either of these compendia, the ingredients shall be suitable for use in the preparation.

3.6 Expiration dating period. The expiration dating period shall be not less than 18 months.

3.7 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.8 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified here, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating

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period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated item, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of syrup produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immedaite container

4.3.1 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay:			
Phenylpropanolamine Hydrochloride	3.3.1.1	S-2	5
Codeine Phosphate	3.3.1.2	S-2	5
Guaifenesin	3.3.1.3	S-2	5 composite
Alcohol	3.3.1.4	S-2	5
pH	3.3.2	S-2	5
Palatability	3.3.3	-	

1/ Where possible, the sample shall be used for two or more tests.

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4.4 Examination. The finished syrup shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Syrup not free from foreign matter.
102	Syrup has crystal formation.
103	Syrup not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

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4.7.1 Assay for phenylpropanolamine hydrochloride, codeine phosphate and guaifenesin.

Column preparation:

Anion column. Place a pledget of glass wool in the bottom of a 1.5 cm (inside diameter) glass column. Add a slurry of Bio-Rex 5 anion exchange resin (or equivalent) until a bed height of about 3 cm remains after the water drains through.

Cation column. Place a pledget of glass wool into the bottom of a 1.0 cm (inside diameter) glass column with a teflon stop-cock at the bottom. Add a slurry of Rexyn 102H cation exchange resin (or equivalent) until a bed height of 11-12 cm remains after the water drains through. Then, wash the column with 100 ml of 1.0 M magnesium sulfate solution and rinse with 50 ml of purified water.

The anion column is then fitted on top of the cation column and a 250 ml reservoir is fitted on top of the cation column and a 250 ml reservoir is fitted on top of the anion column.

Standard preparation:

Phenylpropanolamine hydrochloride. Accurately weigh 375.0 mg of phenylpropanolamine hydrochloride NF reference standard into a 100 ml volumetric flask. Dissolve and dilute to volume with purified water. Pipet 10.00 ml into a 200 ml volumetric flask. Add 150 ml of 0.035 M magnesium sulfate solution and dilute to volume with purified water.

Codeine phosphate. Accurately weigh 100 mg of codeine phosphate USP reference standard into a 100 ml volumetric flask. Dissolve and dilute to volume with purified water. Pipet 10.00 ml into a 100 ml volumetric flask. Add 75 ml of 0.2 N hydrochloric acid and dilute to volume with purified water.

Guaifenesin. Accurately weigh 100 mg of guaiferesin NF reference standard into a 100 ml volumetric flask. Dissolve and dilute to volume with purified water. Pipet 4.00 ml into a 100 ml volumetric flask and dilute to volume with purified water.

Assay procedure:

Sample preparation. Shake the sample and allow the air bubbles to dissipate. Pipet 10.0 ml of sample, using a "To Contain" pipet, into the 250 ml reservoir. Rinse the pipet into the reservoir using purified water. Dilute the sample to 50 ml with purified water and mix.

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Guaifenesin. Place a 200 ml volumetric flask under the cation exchange column to collect the effluent from the initial sample elution and the subsequent washes. After the initial 50 ml containing the diluted sample passes through the column, the column is then rinsed with two 25 ml portions of purified water. The anion column is then removed and the cation column is rinsed with 150 ml of 10 percent methanol (pressure may be used, 1-2 psi). The first 90 ml of the 10 percent methanol rinse is collected with the previous rinses. Discard the remainder of the 10 percent methanol rinse. Remove the 200 ml volumetric flask, dilute to volume with purified water, and mix well. Pipet 4.00 ml of this solution into a 100 ml volumetric flask, dilute to volume with purified water and mix well.

Phenylpropanolamine hydrochloride. Place a second 200 ml volumetric flask under the cation exchange column. Elute the phenylpropanolamine from the column with 150 ml of 0.035 M magnesium sulfate. Remove the flask, dilute to volume with purified water and mix well.

Codeine phosphate. Place a third 200 ml volumetric flask under the cation exchange column. Elute the codeine from the column with 150 ml of 0.2 N hydrochloric acid. Remove the flask, dilute to volume with purified water and mix well.

Spectrophotometric measurement:

Guaifenesin. Using a suitable spectrophotometer, record the absorption spectra of the sample and standard solutions from 300 nm to 240 nm. Measure the absorbance at the wavelength of maximum absorbance and the absorbance at 300 nm. Subtract the absorbance at 300 nm from the absorbance maximum and use this value in the calculation.

Phenylpropanolamine hydrochloride. Using a suitable spectrophotometer, record the absorption spectra of the sample and standard solutions from 280 nm to 225 nm. Measure the absorbance at the wavelength of maximum absorbance using the baseline technique.

Codeine phosphate. Using a suitable spectrophotometer, record the absorption spectra of the sample and standard solutions from 320 nm to 255 nm. Measure the absorbance at the wavelength of maximum absorbance using the baseline technique.

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Calculation:

$$\frac{A_u \times 100}{As} = \text{percent of required amount of ingredient}$$

Where:

A_u = absorbance of sample sclution

As = absorbance of standard solution

4.7.2 Alcohol. Shall be determined by a USP method. As an alternate, the alcohol content may be determined by a suitable gas chromatography method yielding accurate and reproducible results.

4.7.3 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the Palatability Standard 1/, when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the sample shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ Palatability standard is available upon separate request to the contracting officer, Defense Personnel Support Center.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

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5.2.1 Unit of issue. One bottle containing 4 fl oz (118 ml), as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle with child-proof closure considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twenty-four (24) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Forty-eight (48) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Forty-eight (48) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

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5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life markings as specified in MIL-STD-129 shall be shown.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-033-9429	PHENYPROPANOLAMINE HYDROCHLORIDE, CODEINE PHOSPHATE, AND GUAIFENESIN SYRUP, 4 fl oz (118 ml)

Preparing activity:
DLA-DM

Project No. 6505-1922

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-020/4015

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-P-37777(DLA-DM) - PHENYLPROPANOLAMINE HYDROCHLORIDE, CODEINE PHOSPHATE, AND GUAIFENESIN SYRUP

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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